Xueru Wen

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Education

University of Chinese Academy of Sciences, Beijing 2023 – Present

Ph.D. Student in Computer Application Technology; expected 2028; advisor: Prof. Ben He
Chinese Information Processing Laboratory, Institute of Software, CAS

Jilin University, Changchun 2019 – 2023

B.S. Tang Aoqing Honors Program in Science, Computer Science Track

Selected Research

CoVRL: Verifier-free RL [1] First author, ICML 2026

Proposed variational RL coupling prior and posterior sampling to improve exploration and thought–answer consistency. Qwen2.5-7B average accuracy on math and general reasoning rose from 37.8% to 50.2% (+12.4 percentage points), exceeding the strongest compared baseline by 2.3 points.

Reward Model Evaluation [4] First author, ICLR 2025 Spotlight

Showed in synthetic experiments that similar reward-model accuracy can yield different downstream RL outcomes. Analyzed evaluation design and reward overoptimization through the Goodhart effect, exposing the limits of accuracy-based model selection.

Cheems: Chinese Reward Models [5] First author, ACL 2025

Built a Chinese reward-model training and evaluation framework; released CheemsPreference and CheemsBench, evaluated open-source models on Chinese data, and trained Chinese reward models.

RLFH: Self-alignment [6] First author, ACL 2025

Proposed on-policy self-alignment using atomic facts and external knowledge to construct dense token-level rewards without human annotations; reduced hallucination and improved accuracy on benchmarks including HotpotQA.

Experience

RedNote (Ace Top Intern since Jul. 2025) Sep. 2023 – Present

Reward Model Iteration: Iterated RLHF reward models with preference feedback, hallucination detection, and process rewards; evaluated downstream effects and explored scalable oversight mechanisms.

Math Verification Agent: Built an agent with Lean, Python, and natural-language verifiers, then strengthened it with SFT and Agentic RL to improve ProofBench performance.

SWE Coding Agent: Built Docker/Kubernetes sampling pipelines for large-scale trajectory collection and validation; used SFT to evaluate data effectiveness. Implemented TITO, R3, and partial rollout for Agentic RL, improving SWE-bench Verified, Multilingual, and Pro performance.

Automated Posttrain: Built large-scale data synthesis pipelines on cloud GPU clusters, used world models for targeted synthesis, and implemented segment-wise tokenization for training and inference to improve PostTrainBench performance.

Chinese Information Processing Laboratory, ISCAS Mar. 2023 – Aug. 2023

Large Language Model Pre-training

- Added LLaMA training, checkpoint conversion, and numerical alignment to Megatron–DeepSpeed
- Integrated optimizations such as fused kernels and Flash Attention, improving training throughput

Honors and Awards

- CAST Young Talent Support Program, Doctoral Student Special Track 2025
- President's Excellence Award, Chinese Academy of Sciences Aug. 2026

Publications

[1] **Coupled Variational Reinforcement Learning for Language Model General Reasoning**
Xueru Wen, Jie Lou, Yanjiang Liu, Hongyu Lin, Ben He, Xianpei Han, Le Sun, Yaojie Lu, Debing Zhang. ICML 2026

[2] **Decoupling Reasoning and Confidence: Resurrecting Calibration in Reinforcement Learning from Verifiable Rewards**
Zhengzhao Ma, **Xueru Wen**, Boxi Cao, Yaojie Lu, Hongyu Lin, Jinglin Yang, Min He, Xianpei Han, Le Sun. ICML 2026

[3] **Learning from Failures: Correction-Oriented Policy Optimization with Verifiable Rewards**
Mengjie Ren, Jie Lou, Boxi Cao, **Xueru Wen**, Hongyu Lin, Xianpei Han, Le Sun, Xing Yu, Yaojie Lu. COLM 2026

[4] **Rethinking Reward Model Evaluation: Are We Barking up the Wrong Tree?**
Xueru Wen, Jie Lou, Yaojie Lu, Hongyu Lin, Xing Yu, Xinyu Lu, Ben He, Xianpei Han, Debing Zhang, Le Sun. ICLR 2025 Spotlight

[5] **Cheems: A Practical Guidance for Building and Evaluating Chinese Reward Models from Scratch**
Xueru Wen, Jie Lou, Zichao Li, Yaojie Lu, Xing Yu, Yuqiu Ji, Guohai Xu, Hongyu Lin, Ben He, Xianpei Han, Le Sun, Debing Zhang. ACL 2025

[6] **On-Policy Self-Alignment with Fine-grained Knowledge Feedback for Hallucination Mitigation**
Xueru Wen, Jie Lou, Xinyu Lu, Yuqiu Ji, Xinyan Guan, Yaojie Lu, Hongyu Lin, Ben He, Xianpei Han, Debing Zhang, Le Sun. ACL 2025

[7] **Transferable Post-training via Inverse Value Learning**
Xinyu Lu, **Xueru Wen**, Yaojie Lu, Bowen Yu, Hongyu Lin, Haiyang Yu, Le Sun, Xianpei Han, Yongbin Li. NAACL 2025

[8] **AutoAlign: Get Your LLM Aligned with Minimal Annotations**
Xinyu Lu, Dong Xu, Chunkang Zhang, Xinyan Guan, Junxiang Wang, Qingyu Zhang, Pengbo Wang, Yingzhi Mao, Hao Xiang, **Xueru Wen**, Zichao Li, Yaojie Lu, Hongyu Lin, Le Sun, Xianpei Han. ACL 2025

[9] **Critic-CoT: Boosting the Reasoning Abilities of Large Language Model via Chain-of-Thought Critic**
Xin Zheng, Jie Lou, Boxi Cao, **Xueru Wen**, Yuqiu Ji, Hongyu Lin, Yaojie Lu, Xianpei Han, Debing Zhang, Le Sun. ACL 2025

[10] **The Devil Is in the Details: Tackling Unimodal Spurious Correlations for Generalizable Multimodal Reward Models**
Zichao Li, **Xueru Wen**, Jie Lou, Yuqiu Ji, Yaojie Lu, Xianpei Han, Debing Zhang, Le Sun. ICML 2025

[11] **Offline Pseudo Relevance Feedback for Efficient and Effective Single-Pass Dense Retrieval**
Xueru Wen, Xiaoyang Chen, Xuanang Chen, Ben He, Le Sun. SIGIR 2023

[12] **End-to-End Entity Detection with Proposer and Regressor**
Xueru Wen, Changjiang Zhou, Haotian Tang, Luguang Liang, Hong Qi, Yu Jiang. Neural Processing Letters 2023

[13] **Type-supervised Sequence Labeling Based on the Heterogeneous Star Graph for Named Entity Recognition**
Xueru Wen, Changjiang Zhou, Haotian Tang, Luguang Liang, Yu Jiang, Hong Qi. arXiv 2022

Open-Source Tools

- **Skynet**: A native macOS agent workspace with local/SSH sessions, queued follow-ups, and code review.
- **Python Hot Patch Skill**: Live CPython hot-patching with preserved state, safe-point execution, and verification.
- **Wait Skill**: Event-driven agent waiting and coordination, with monitoring, timed loops, and dependency workflows.
- **JSONLine Viewer**: A VS Code extension for formatted JSONL/NDJSON previews and indexed navigation of large files.